A Metacommunity Model and its Diffusion Approximation

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Based on the preprint "Feller Property and Absorption of Diffusions for Multi-Species Metacommunities", a joint work with Benoît Henry.

The Paper

Terminology

- **Patch**: A given area with finite hosting capacity.
- **Population**: Individuals of a single species in a patch.
  Metapopulation: A set of populations spread across several patches.
- **Community**: Interacting individuals from different species in a patch.
  Metacommunity: A set of interacting communities spread across several patches.

A Finite Population Model

Generalisation of a metacommunity model [1]:

- 2 species,
- capacities: respectively, $N_1 = \lfloor d_1 N \rfloor, \dots, N_m = \lfloor d_m N \rfloor$,
- 2 patches $\longrightarrow$ m patches,
- linear dispersal $\longrightarrow$ general dispersal.

Let the Markov chain $(Y_n)_{n\in\mathbb{N}}$ model the evolution of the species 1 density.

Transition from a generation to the next one:

$$Y_n \xrightarrow{\text{Wright–Fisher}} Y_{n+1}^{WF} \xrightarrow{\Phi_N} Y_{n+1}$$

1. **Reproduction phase (stochastic)**: neutral Wright–Fisher *independently* on each patch.

   Every individual in a generation is the offspring of a parent chosen uniformly and independently at random from the previous generation.

2. **Migration phase (deterministic)**:

   $$Y_{n+1} = \Phi_N(Y_{n+1}^{WF}),$$

   where $\Phi_N : [0,1]^m \to [0,1]^m$.

We obtain a Markov chain $(Y_n)_{n\in\mathbb{N}}$ with state space $[0,1]^m$. We then *poissonise* the Markov chain with rate N to obtain a Markov jump process $(X_t^N)_{t\in\mathbb{R}_+}$. Its infinitesimal generator is a bounded operator on $\mathcal{C}([0,1]^m)$ given by

$$L_N f = N(B_N(f \circ \Phi_N) - f), \quad f \in \mathcal{C}([0,1]^m),$$

where B_N is the m-dimensional Bernstein operator on $\mathcal{C}([0,1]^m)$.

Hypotheses on the Model

1. **Conservation of metapopulation size through migration**:

   $$\underbrace{N_1\Phi_N^{(1)}(x) + \cdots + N_m\Phi_N^{(m)}(x)}_{\text{Metapopulation size } \textit{after} \text{ migration}} = \underbrace{N_1 x_1 + \cdots + N_m x_m}_{\text{Metapopulation size } \textit{before} \text{ migration}}.$$

2. **Net migration approximation**:

   $$\lim_{N\to\infty} \sup_{x\in[0,1]^m} |N(\Phi_N(x) - x) - b(x)| = 0,$$

   for some $b \in \mathcal{C}^2([0,1]^m, \mathbb{R}^m)$.

Properties on b

- b is inward-pointing,
- $\sum_{i=1}^m d_i b_i = 0$,
- $b(0,\dots,0) = b(1,\dots,1) = 0$.

Scaling Limit: What happens when $N \to \infty$?

Notice that for any $f \in \mathcal{C}^2([0,1]^m)$,

$$\lim_{N\to\infty} L_N f = Lf, \tag{1}$$

uniformly on $[0,1]^m$, where

$$Lf = \frac{1}{2}\sum_{i=1}^m \frac{x_i(1-x_i)}{d_i}\frac{\partial^2 f}{\partial x_i^2} + \sum_{i=1}^m b_i(x)\frac{\partial f}{\partial x_i}. \tag{2}$$

L is elliptic on $[0,1]^m$, but degenerates on the boundary.

- L is the differential operator of the Stochastic Differential Equation (SDE)

  $$X_t = X_0 + \int_0^t \sigma(X_s)\,dB_s + \int_0^t b(X_s)\,ds, \tag{3}$$

  where $\sigma(x) = \operatorname{diag}\left[\sqrt{\frac{x_1(1-x_1)}{d_1}}, \dots, \sqrt{\frac{x_m(1-x_m)}{d_m}}\right]$.
- **Characterisation**: if weak existence and uniqueness holds for (3), is $\{(f, Lf),\ f \in \mathcal{C}^2([0,1]^m)\}$ the graph of its infinitesimal generator?

Theorem (Existence and Core for its Generator)

The operator L defined in (2) on $\mathcal{C}^2([0,1]^m)$ is closable, and its closure generates a Feller semigroup.

Interesting because...

- We can use (1) to prove the weak convergence of (X^N).
- The generator of the diffusion is characterised *without specifying boundary conditions*.

The proof is based on [2].

Finite Extinction time?

Consider the limit diffusion described by SDE (3). Let T_{ext} be the extinction time of one species, *i.e.* the hitting time of the absorbing states $\{(0,\dots,0),(1,\dots,1)\}$.

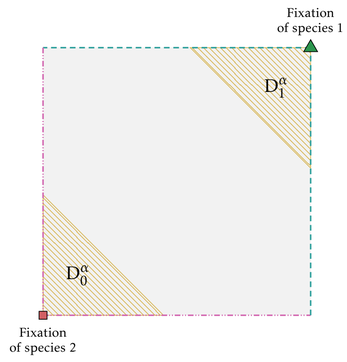
Theorem (Extinction)

Assume that b is strictly inward-pointing except at $(0,\dots,0)$ and $(1,\dots,1)$. Then for any $x \in [0,1]^m$, $\mathbb{P}_x[T_{ext} < \infty] = 1$.

Idea of the Proof

Introduce:

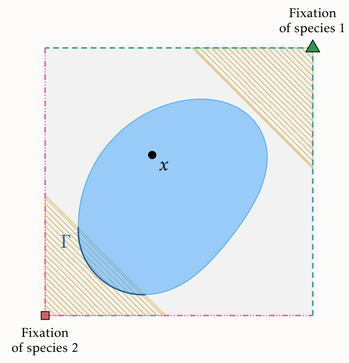
- $\varphi_d : x \mapsto \frac{1}{d_1\cdots d_m}\sum_{i=1}^m x_i$,
- $D_0^\alpha = \{\varphi_d < \alpha\}$ and $D_1^\alpha = \{\varphi_d > \varphi_d(1_m) - \alpha\}$, $\alpha \le \min(1, \varphi_d(1_m))$,
- T_α the hitting time of $D_0^\alpha \cup D_1^\alpha$,
- $h_\alpha : x \mapsto \mathbb{P}_x[T_\alpha < \infty]$,
- $h : x \mapsto \mathbb{P}_x[T_{ext} < \infty]$.



Outline:

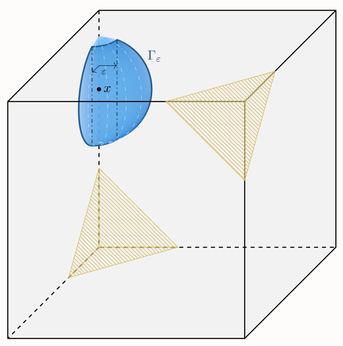
1. $\forall x \in [0,1]^m,\ h_\alpha(x) > 0$.
   Starting from any point x, the process attains $D_0^1 \cup D_1^\alpha$ with positive probability.
2. $h_\alpha \equiv 1$.
   - 2.1 $h_\alpha(X)$ is a supermartingale,
   - 2.2 h_α is lower semicontinuous.
3. $h \equiv 1$.
   - 3.1 Markov property with time T_α,
   - 3.2 $h(X)$ is a martingale.

Point 1 (Sketch)



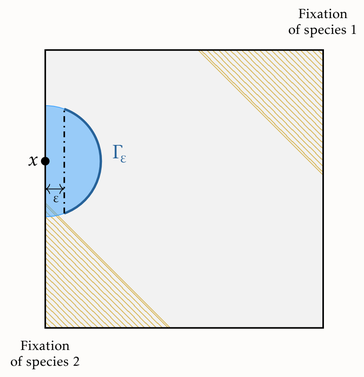
Case 1: $x \in]0,1[^m$.

There exists an open subset D containing x. L is uniformly elliptic on D, and considering the right Dirichlet problem, we prove that $\mathbb{P}_x[T_\alpha < \infty] > 0$.



Case 2: $x \in \partial([0,1]^m)$.

We prove that the process hits a face of higher dimension in finite time with positive probability by hitting the boundary of a ball centred at the initial state.



Recursively, we prove that it will reach the interior in finite time with positive probability.

References

[1] G. Delvoye, O. Goubet, and F. Paccaut. "Comparison Principles and Applications to Mathematical Modelling of Vegetal Meta-Communities". In: *Mathematics in Engineering* 4 (2021), pp. 1–17. doi: 10.3934/mine.2022035.

[2] S. N. Ethier. "A Class of Degenerate Diffusion Processes Occurring in Population Genetics". In: *Communications on Pure and Applied Mathematics* 29.5 (1976), pp. 483–493. doi: 10.1002/cpa.3160290503.